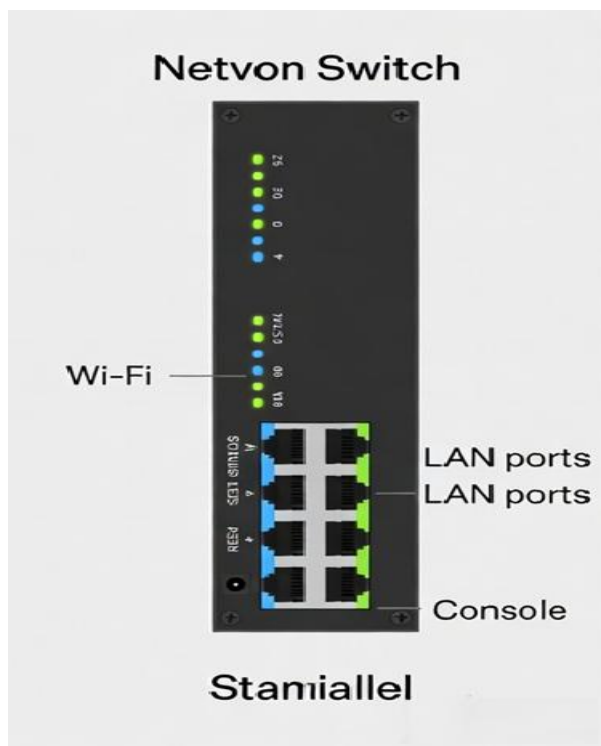


DAY-2

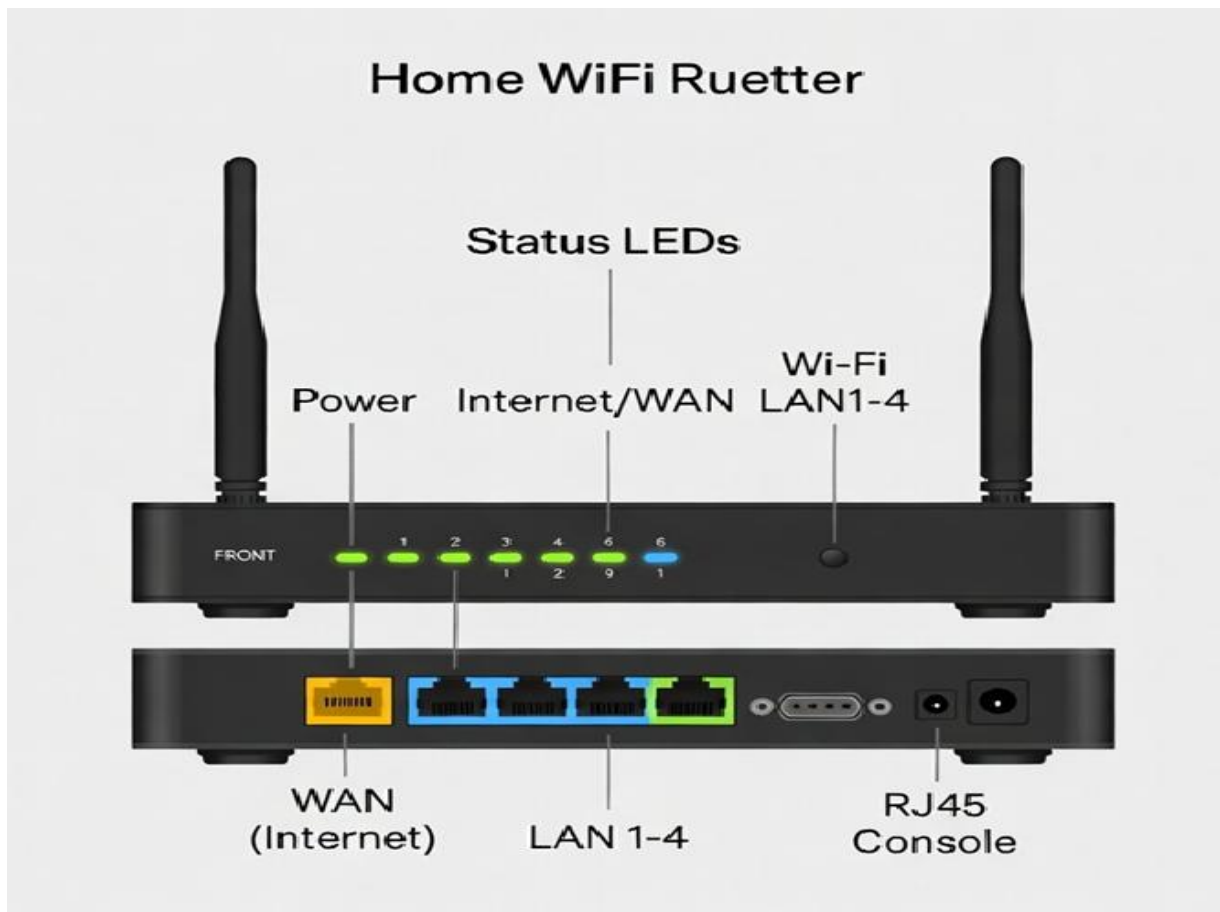
Router and switches theory

1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

- Network switch



- Wi-fi router



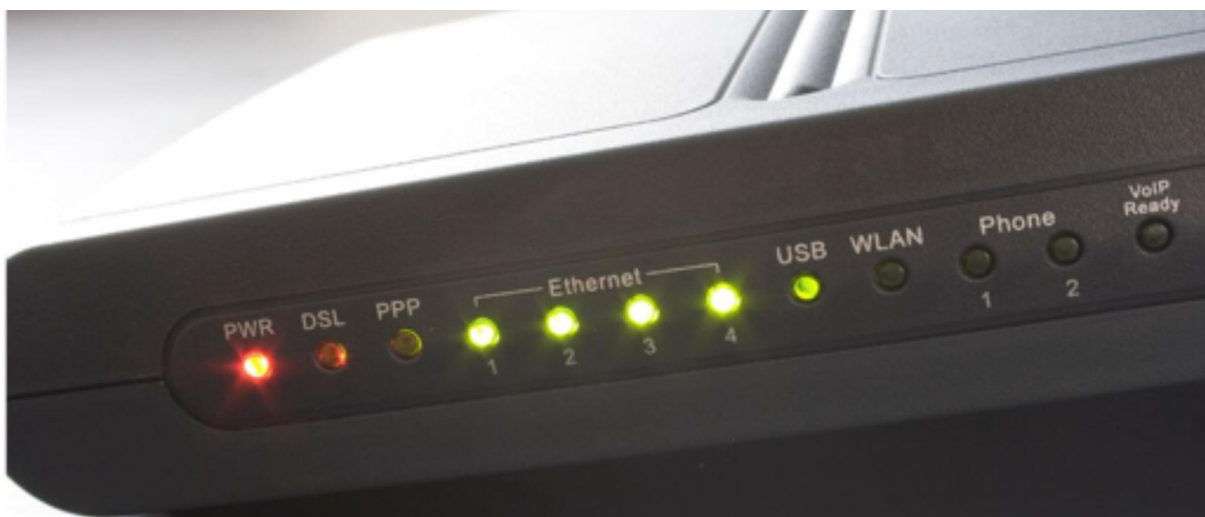
2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

<i>Aspect</i>	<i>Router</i>	<i>Switch</i>
<i>Main function</i>	<i>Connects different networks, such as a home LAN to the Internet, and chooses the best path for packets.</i>	<i>Connects devices within the same local network, such as computers, printers, and servers.</i>
<i>Address used</i>	<i>Uses IP addresses to determine where network packets should go, maintaining a routing table.</i>	<i>Uses MAC addresses to forward Ethernet frames to the correct device, maintaining a MAC-address table.</i>
<i>Traffic handled</i>	<i>Separates networks and forwards traffic between different IP networks, limiting broadcasts between them.</i>	<i>Efficiently forwards traffic inside a LAN, sending frames only to the relevant port when it knows the destination MAC address.</i>

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

Router

- *Power port: Usually on the rear panel, labeled Power, DC IN, or shown with a power symbol. It connects to the router's power adapter and supplies electricity.*
- *Ethernet ports: The rear typically has one WAN/Internet port and several LAN ports. The WAN port connects to the modem or outside network; LAN ports connect local devices such as computers or switches.*
- *LED indicators: Usually on the front or top of the router. Common LEDs include Power, Internet/WAN, Wi-Fi, and LAN indicators. They show whether the router is powered on and whether connections or data activity are present.*



Switch

- *Power port: Usually at one end or on the rear, labeled Power or DC IN. It connects to the switch's power adapter.*
- *Ethernet ports: The front row contains multiple RJ-45 Ethernet ports, commonly numbered 1–5 or 1–8. Each port connects devices on the same local network; on an ordinary switch, any port can generally be used for a computer, router, printer, or access point.*
- *LED indicators: Small LEDs are located beside or above the Ethernet ports. A Power LED shows that the switch is on, while individual Link/Activity LEDs show that a device is connected and may blink when data is transmitted or received.*



4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

You'd connect the router first to the ISP line, then use a switch to connect all 8 gaming PCs.

Why the router comes first

The ISP (Internet Service Provider) gives you a single public IP address and expects to talk to one device that will act as your network's gateway. A router is designed for exactly this role:

- It connects your local network (LAN) to the internet (WAN).*
- It performs NAT, so all 8 PCs can share that single public IP.*
- It provides DHCP, firewalling, and basic traffic management—critical in a gaming café where you want stability and some control over bandwidth.*

If you put a switch directly on the ISP line, each PC would try to get its own public IP from the ISP, which usually won't work

(most ISPs don't give out 8 public IPs for a small café plan) and would also expose every PC directly to the internet with no central firewall.

How to connect the 8 gaming PCs

A typical, clean setup for your small IPL gaming café:

1. ISP line → Router WAN port

- The fiber/DSL/cable from the ISP plugs into the router's WAN/Internet port.*

2. Router LAN port → Switch

- Use an Ethernet cable from one of the router's LAN ports to any port on a managed or unmanaged Gigabit switch (8-port or larger).*

3. Switch → 8 gaming PCs

- Connect each of the 8 gaming PCs to the switch using Ethernet cables.*
- For low-latency IPL streaming and multiplayer matches, wired connections are strongly preferred over Wi-Fi.*

Optionally, if your router has enough LAN ports (e.g., 8), you could plug the PCs directly into the router, but a dedicated switch gives you:

- Better port density and room to expand.*

- *Potentially better performance and features (QoS, VLANs) if you use a managed switch.*

Justification in scenario terms

Imagine your café “Café Sixer” opening for IPL:

- *The router is the “front desk” talking to the ISP, handling one public IP, assigning internal IPs (e.g., 192.168.1.x) to all PCs, and protecting them with a firewall.*
- *The switch is the “internal wiring hub” that lets all 8 gaming stations talk to each other at full Gigabit speeds for local multiplayer matches while sharing the router’s internet connection for live IPL streams and online games.*

This order—ISP → Router → Switch → PCs—is the standard, reliable topology for any small business or gaming café network.